Deep Learning Project 4: LoRA Style-Tuning

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Base model: runwayml/stable-diffusion-v1-5

Training data: style_imgs/512

Evaluation prompt: a busy market, in <sks> style

Method

- Added the style token <sks> to the Stable Diffusion tokenizer and trained its embedding row.
- Applied LoRA adapters to both UNet attention projections and CLIP text encoder attention projections.
- Trained on all 843 supplied style images with fixed Florence-2 captions.
- Added 60 deterministic images generated by unmodified SD 1.5: broad markets, market characters, and people scenes.
- No assignment-PDF image, external dataset, or adapter-generated image was used.
- Fixed the complete 2,000-draw source and augmentation schedule before loading the model.
- Used deterministic per-draw seeds for crops, flips, prompts, VAE latents, diffusion noise, and timesteps.
- Selected step 250 after long-horizon checkpoint reviews and blind market, portrait, and two-person holdouts.
- Used Min-SNR gamma 5, adapter-off frozen-base preservation, and a token anchor.
- Stored the UNet LoRA, text encoder LoRA, and learned token embedding in one safetensors file.

Hyperparameters

- Token: <sks>
- Resolution: 512
- LoRA ranks: UNet 16; text encoder 4
- Selected checkpoint: step 250; LRs UNet 7e-5, text 2.5e-6, token 1e-5
- Batch 1; accumulation 4; fp16; seed 2202; random crop and horizontal flip
- Effective data weights: 86% supplied, 5% broad markets, 5% market characters, 4% people
- Min-SNR gamma 5; preservation weight 0.65; token anchor 0.05
- LoRA dropout 0.05; caption dropout 0.08; cosine schedule; 50 warmup steps
- Registered max 500 steps; visually selected checkpoint at step 250
- Every draw logged; immutable adapters and resume state saved every 25 steps
- Weights: lora_out\pytorch_lora_weights.safetensors

Notes

- The evaluation script renders at least three images and does not expose a style-strength slider.
- Supplied-only training transferred style well but produced flatter market scenes and simpler crowd faces.
- Original-SD market preservation improved stall geometry, crowd separation, and target-prompt reliability.
- An 8% supplied-image face-crop share improved close portraits but not small crowd faces consistently.
- The original cosine path improved through step 250 and then converged without further visual gains through step 500.
- A stronger-preservation continuation peaked at cumulative step 225 but lost the final blind comparison.
- Across 20 blind holdout seeds, step 250 won 11, the preserved continuation won 6, and step 150 won 3.
- All ten final market holdouts were populated; distant faces remain limited at 512x512.
- The reproduced initialization and continuation checks matched all 353 tensors at verified boundaries.

Generated Samples

Prompt: a busy market, in <sks> style

adapter_00.png

adapter_01.png

adapter_02.png

adapter_03.png

adapter_04.png

adapter_05.png